

# Arnab Bhattacharjee

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## Research Interests

Human-Computer Interaction (HCI), Human-AI Interaction, LLM Agents and Agentic AI, Systems

## Education

**Bangladesh University of Engineering and Technology (BUET)**

February 2020 - March 2025

*B.Sc. in Computer Science and Engineering*

CGPA: 3.88 / 4.00

- Thesis: *Evaluating the Multilingual Capabilities of LLM-based Web UI Agent*, supervised by Dr. Rifat Shahriyar
- Dean's List Scholarship (Level 1 and Level 2)

**Notre Dame College, Dhaka**

June 2017 - May 2019

*Higher Secondary Certificate (HSC)*

GPA: 5.00 / 5.00

## Research Experience

**Human-AI Co-Creation for Technical Lecture Content** | *HCI, Agentic AI*

2026

- Investigated how computer science educators author technical lecture content with generative AI tools.
- In a formative study, conducted semi-structured interviews with university educators and analyzed them using reflexive thematic analysis to identify their challenges and derive design goals.
- Guided by these goals, built an agentic pipeline that generates technical lecture content under the educator's direction.
- Conducted a qualitative user evaluation through focus group discussions (FGDs) with educators.

**Language Effects on LLM Web Agents** | *LLM Agents, Multilingual Evaluation*

- Evaluated an HTML-based LLM web agent across several widely spoken languages in two settings: a controlled offline benchmark in which the page structure stays fixed and only the language changes, and live task execution on real localized websites.
- Quantified how much of the agent's observation actually changes with the language, alongside task performance.
- Found that performance stayed close to English even when most of the page text was rewritten, while non-English pages consumed a substantially larger share of the model's context budget.

**Evaluating the Multilingual Capabilities of LLM-based Web UI Agent** | *Undergraduate Thesis, BUET*

2024 - 2025

- Developed a multilingual web agent that navigates web pages and performs actions to complete complex, multi-step tasks given by a user.
- Fine-tuned a multilingual model (*mT5-base*) on an extended Mind2Web dataset, and built web agents on three frontier LLMs (Gemini-2.0-Flash, GPT-4o-mini, DeepSeek-V3) to evaluate their zero-shot performance.
- Built a VLM-based automated evaluation system for web agents.
- Supervised by Dr. Rifat Shahriyar, Professor, BUET.

## Poster

Abir Muhtasim\*, **Arnab Bhattacharjee\***, Abhik Bhattacharjee, Faria Huq, Rifat Shahriyar. *How well do LLM Web Agents work for non-English Language?* LLM Agent Workshop, Carnegie Mellon University, Pittsburgh, PA, USA. May 2, 2024. Poster, Extended Abstract. (\*Equal contribution)

## Work Experience

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### United International University (UIU)

July 2025 - Present

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

- Courses taught: Data Structures and Algorithms, Discrete Mathematics, Object-Oriented Programming, Operating Systems, Digital Logic Design, Network Security, and Social, Environment and Engineering Ethics.
- Course coordinator for Microprocessor and Microcontroller (theory and lab), including embedded systems projects with Arduino and Raspberry Pi.

### Samsung Research and Development Institute, Bangladesh

May 2025 - July 2025

Software Engineer

Dhaka, Bangladesh

- Developed an Android application (Kotlin, Android Studio) for an intelligent pen that writes on any surface, with the strokes appearing on the Android device automatically.

## Projects

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### Machine Learning Assisted Gene Expression Restoration | *Machine Learning, Bioinformatics*

[GitHub](#)

- Explored nonlinear models for identifying the transcription factors that restore gene expression from a source cell state to a target state, extending the linear GAGER approach. Nonlinear models often reduced expression differences using fewer transcription factors across several scRNA-seq datasets.

### Project Tokkhok | *React, FastAPI, PostgreSQL, Supabase, RAG*

[Demo](#) | [GitHub](#)

- AI-assisted learning platform for Bengali-speaking students: chat with an uploaded knowledge base (retrieval over a vector database) in Bangla, Banglish (Bangla typed in English letters), or audio, and a smart note editor that converts Banglish notes to proper Bangla.

### Buldr-Bangladesh | *React, Spring Boot, PostgreSQL, Tailwind CSS, Firebase Auth*

[GitHub](#)

- Full-stack platform for engineers to learn (AI chatbot, smart notes, summarizing and paraphrasing tools), showcase projects, sell projects through a marketplace with order tracking, and get auto-generated portfolios.

### Deepfake Image Detector Robust Against Adversarial Attacks | *PyTorch, ViT, YOLOv11*

[GitHub](#)

- Detector for real and AI-generated images (CIFAKE). Feature dropping and augmentation reached up to 80% accuracy on adversarial data, and custom adversarial noise and dropout layers improved robustness.

### TechWorld | *Node.js, Express.js, PostgreSQL, Oracle PL/SQL*

[GitHub](#)

- E-commerce website for PCs with complex SQL and PL/SQL operations. Core feature: PC-BUILDER for assembling a custom PC from compatible parts.

## Technical Skills

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**HCI Methods:** Semi-structured interviews, reflexive thematic analysis, focus group discussions, formative studies

**AI and LLMs:** Agentic LLM pipelines, RAG, LLM-as-judge evaluation, web agents, PyTorch, fine-tuning transformer models

**Languages:** Python, C, C++, Java, JavaScript, Kotlin, SQL, Bash

**Web and Databases:** React, Tailwind CSS, Node.js, Express.js, Spring Boot, FastAPI, PostgreSQL, Supabase, Oracle DB

**Tools:** Playwright, Android Studio, Postman, Swagger, LaTeX

## Achievements and Test Scores

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- Champion, ITVerse 2023 "TechTales" Project Showcase Competition, University of Dhaka
- Finalist, BUET CSE Fest Capture the Flag (CTF) Competition 2023
- Dean's List Scholarship, BUET (Level 1 and Level 2)
- IELTS: 7.5 overall (Speaking 7.0)
- Competitive programming: 600+ problems solved, Codeforces max rating 1519 ([hay\\_re\\_manosh](#))

## References

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Available upon request.